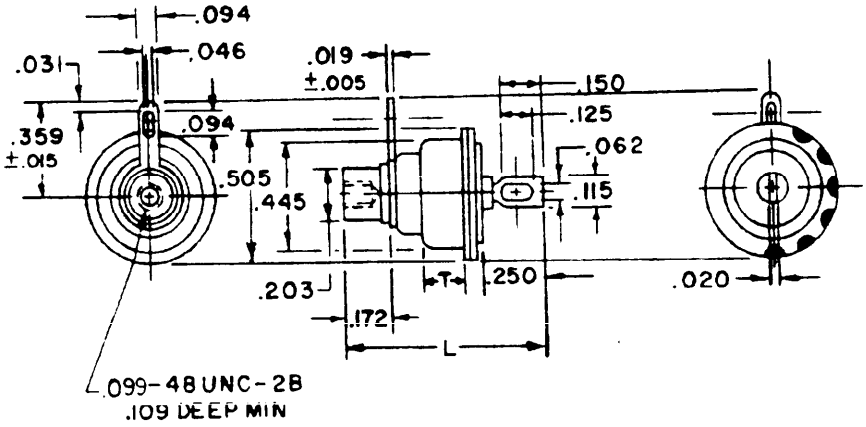
MIL-C-10950/9C
12 June 1978
SUPERSEDING
MIL-C-10950/9B
8 July 1968

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC, BUTTON STYLE (STANDOFF, HERMETIC SEAL) STYLES CB65, CB66, AND CB67

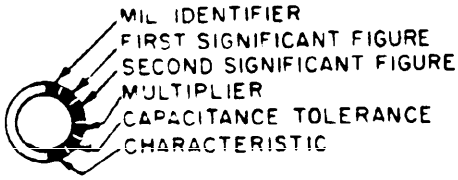
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-10950.



INCHES	MM
.005	.13
.015	.38
.019	.48
.020	.51
.031	.79
.046	1.17
.062	1.57
.094	2.39
.099	2.51
.109	2.77
.115	2.92
.125	3.18
.150	3.81
.172	4.37
.203	5.16
.250	6.35
.359	9.12
.445	11.30
.505	12.83

WITH TERMINAL ASSEMBLY P



COLOR-CODE MARKING

(SEE NOTE 4)

Dimensions		
Style	T±.015(.38)	L±.045(1.14)
CB65	.074 (1.88)	.666 (16.92)
CB66	.100 (2.54)	.692 (17.58)
CB67	.178 (4.52)	.770 (19.56)



CIRCUIT DIAGRAM

NOTES:
1. All dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is .xxx ±.010 (.25 mm).
4. Color-coded section shall not exceed 240°.
5. Ferrous material may be used.

FIGURE 1. Capacitors, (standoff, hermetic seal), styles CB65, CB66, and CB67.

Ⓒ denotes changes

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T14:39Z
Check the source to verify that this is the currentversion before use.

TABLE I. Styles CB65, CB66, and CB67.

Part number 1/	Capacitance	DC rated voltage	Characteristic available at capacitance tolerance				Terminal assembly
			+1% (F)	+2% (G)	+5% (J)	+10% (K)	
	pF	Volts					
CB65PB050K	5	500				B	P
CB65PD100K	10	500				D	P
CB65PD150K	15	500				D	P
CB65PD200J	20	500			D		P
CB65PD220-	22	500			D	D	P
CB65PD330-	33	500			D	D	P
CB65PD470-	47	500			D	D	P
CB65P-680-	68	500	D,F	D,F	D,F	D	P
CB65P-750-	75	500	D,F	D,F	D,F	-	P
CB65P-101-	100	500	D,F	D,F	D,F	D	P
CB65P-111-	110	500	D,F	D,F	D,F	-	P
CB65P-151-	150	500	D,F	D,F	D,F	D	P
CB65P-221-	220	500	D,F	D,F	D,F	D	P
CB65P-331-	330	500	D,F	D,F	D,F	D	P
CB65P-471-	470	500	D,F	D,F	D,F	D	P
CB66P-561-	560	500	D,F	D,F	D,F	D	P
CB66P-681-	680	500	D,F	D,F	D,F	D	P
CB66P-821-	820	500	D,F	D,F	D,F	D	P
CP66P-102-	1,000	500	D,F	D,F	D,F	D	P
CB67P-122-	1,200	500	D,F	D,F	D,F	D	P
CB67P-152-	1,500	500	D,F	D,F	D,F	D	P
CB67P-182-	1,800	500	D,F	D,F	D,F	D	P
CB67P-222-	2,200	500	D,F	D,F	D,F	D	P
CB67P-242-	2,400	500	D,F	D,F	D,F	D	P

1/ Complete part number will include additional symbols to indicate characteristic, and capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See figure 1.
Case: See figure 1.
Terminals: See figure 1.
Capacitance value: See table I.
Capacitance tolerance: See table I.
DC rated voltage: 500 volts.
Dissipation factor: See figure 1 of MIL-C-10950.
Characteristics: See table I.

Symbol	Temperature coefficient	Capacitance drift
B	Not specified	Not specified
D	-100 to +100	0.3 percent or 0.3 pF whichever is greater.
F	-60 to +60	± (0.1 percent + 0.1 pF).

Operating temperature range: -55° to +150°C.

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:39Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
200 percent of rated dc voltage for not less than 1 second, nor more than 5 seconds.

Barometric pressure: Method 105 of MIL-STD-202, condition C (70,000 ft.)
Test potential - 125 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.
Surge current shall not exceed 5 milliamperes.

Insulation resistance: Method 302 of MIL-STD-202, condition A or B.
At 25°C - 100,000 megohms minimum.
At 150°C - 15,000 megohms minimum.

Capacitance: Method 305 of MIL-STD-202.
Measurement frequency:
Capacitance values of 1000 pF or less - 1 MHz ±100 kHz.
Capacitance values of greater than 1000 pF - 1 kHz ±100 Hz.

Shock medium impact: Method 205 of MIL-STD-202, condition C (50 G)

Vibration high frequency: Method 204 of MIL-STD-202, condition B.

Moisture resistance: Test method 106 of MIL-STD-202.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.
Dissipation factor - Initial requirement.

(C) Thermal shock and immersion:
Thermal shock: Test method 107 of MIL-STD-202, condition A, except temperature for step 3 shall be $150^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
Immersion: Test method 104 of MIL-STD-202, condition A.
Insulation resistance: Not less than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Life: 2000 ±12 hours at $150^{\circ}{}^{+3^{\circ}}_{-0^{\circ}}$ C.
Insulation resistance: Greater than 100,000 megohms.
Capacitance change: Within 1 percent of the initial value or 0.5 pF, whichever is greater.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Torque: In accordance with MIL-C-10950.
Mounting - normal mounting means.
Load - 4 ounce-inches.

NOTES:
1. Conversion of dash numbers to part numbers is as shown in table II.
(C) 2. Characteristic E has been replaced by characteristic F.

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:39Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion.

Dash number in MIL-C-10950/9A		Converted to part number 1/ in
Characteristic D	Characteristic E	MIL-C-10950/9B and 9C
0001	0002	CB65P-680F
0003	0004	CB65P-101F
0005	0006	CB65P-151F
0007	0008	CB65P-221F
0009	0010	CB65P-331F
0011	0012	CB65P-471F
0013	0014	CB66P-681F
0015	0016	CB66P-821F
0017	0018	CB66P-102F
0019	0020	CB67P-122F
0021	0022	CB67P-152F
0023	0024	CB67P-182F
0025	0026	CB67P-222F
0027	0028	CB67P-242F
0029	0030	CB65P-680G
0031	0032	CB65P-101G
0033	0034	CB65P-151G
0035	0036	CB65P-221G
0037	0038	CB65P-331G
0039	0040	CB65P-471G
0041	0042	CB66P-681G
0043	0044	CB66P-821G
0045	0046	CB66P-102G
0047	0048	CB67P-122G
0049	0050	CB67P-152G
0051	0052	CB67P-182G
0053	0054	CB67P-222G
0055	0056	CB67P-242G
0057		CB65PD220J
0058		CB65PD330J
0059		CB65PD470J
0060	0061	CB65P-680J
0062	0063	CB65P-101J
0064	0065	CB65P-151J
0066	0067	CB65P-221J
0068	0069	CB65P-331J
0070	0071	CB65P-471J
0072	0073	CB66P-681J
0074	0075	CB66P-821J
0076	0077	CB66P-102J

1/ See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:39Z
Check the source to verify that this is the current version before use.

TABLE II. Dash number to part number conversion (continued).

Dash number in MIL-C-10950/9A		Converted to part number 1/ in MIL-C-10950/9B and 9C
Characteristic D	Characteristic E	
0078	0079	CB67P-122J
0080	0081	CB67P-152J
0082	0083	CB67P-182J
0084	0085	CB67P-222J
0086	0087	CB67P-242J
0088		CB65PB050K
	0089	CB65PD100K
	0090	CB65PD150K
	0091	CB65PD220K
	0092	CB65PD330K
	0093	CB65PD470K
	0094	CB65PD680K
	0095	CB65PD101K
	0096	CB65PD151K
	0097	CB65PD221K
	0098	CB65PD331K
	0099	CB65PD471K
	0100	CB66PD681K
	0101	CB66PD821K
	0102	CB66PD102K
	0103	CB67PD122K
	0104	CB67PD152K
	0105	CB67PD182K
	0106	CB67PD222K
	0107	CB67PD242K

1/ The complete part number shall include symbols for terminal assembly and characteristic, as applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - AR
Navy - SH
Air Force -17, 85
DLA - ES

User activities:
Navy - AS, CG, OS

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5910-1264-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:39Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316



COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-C-10950/9C 12 JUNE 1978

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 20261O-17T14:39Z
Check the source to verify that this is the current version before use.